

State and Parameter Estimation for Affine Nonlinear Systems

CDC 2023

Tochukwu Elijah Ogri ¹ Zachary I. Bell ² **Rushikesh Kamalapurkar** ¹

¹Oklahoma State University, Oklahoma, USA

²Air Force Research Laboratory, Florida, USA

December 13, 2023



Outline

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Motivation

- ▶ Control techniques such as adaptive optimal control rely on accurate models.
- ▶ State-space system identification typically requires full state measurements.
- ▶ Adaptive observers do not generally focus on parameter convergence. Those that do require existence of coordinate transformations that transform the system in a form where the regressor is a function of the output and the input, but not the state (Pyrrkin et al., 2019).
- ▶ This paper integrates:
 - (a) multiplier matrix approach for Luenberger state observer design,
 - (b) an event based implementation of concurrent learning (CL) for estimating parameters, and
 - (c) an algorithm for refining parameters estimates with improved state estimatesinto a state and parameter estimation framework.

Problem Formulation

- ▶ Nonlinear dynamical system: $\dot{x} = Y(x)\theta + g(x)u$, $y = Cx$
- ▶ Main objectives:
 - ▶ Develop a state observer for estimation of x using input u and output y .
 - ▶ Develop a parameter estimation scheme to provide estimates $\hat{\theta}$.

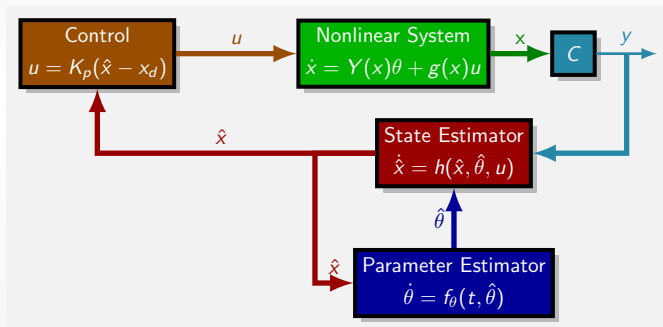


Figure: State and Parameter Estimator

Problem Formulation

Assumption 1: Bounded Jacobians

The functions Y and g are known, their derivatives exist on the compact set $\mathcal{C} \subset \mathbb{R}^n$ containing the origin and satisfy the element-wise bounds

$$(K_{y_1})_{j,k} \leq \left(\frac{\partial(Y(x))_{j,i}}{\partial(x)_k} \right) \theta_i \leq (K_{y_2})_{j,k},$$
$$(K_{g_1})_{j,k} \leq \left(\frac{\partial(g(x))_{j,l}}{\partial(x)_k} \right) u_l \leq (K_{g_2})_{j,k},$$

for all $x \in \mathcal{C}$, $u \in \mathcal{U}$, $\theta \in \Theta$, $i = 1, \dots, p$, $j, k = 1, \dots, n$, and $l = 1, \dots, m$.

Assumption 2: Bounded Parameters

The unknown parameters θ belong to a known convex and compact set Θ .

State Observer Design

- Using the derivative bounds in Assumption 1, a state observer with three correction terms is designed as:

$$\dot{\hat{x}} = A\hat{x} + F_{\theta}[\hat{x} + l_1(y - C\hat{x}), \hat{\theta}] + G_u[\hat{x} + l_2(y - C\hat{x}), u] + L(y - C\hat{x})$$

where $A = K_{y1} + K_{g1}$, $F_{\theta}(x, \theta) = -K_{y1}x + Y(x)\theta$, $G_u(x, u) = -K_{g1}x + \sum_{i=1}^N g_i(x)(u)_i$, $l_1 \in \mathbb{R}^{n \times q}$, $l_2 \in \mathbb{R}^{n \times q}$ and $L \in \mathbb{R}^{n \times q}$ are observer gains.

- The observer error is defined as

$$\tilde{x} = x - \hat{x}$$

and the observer error dynamics can be expressed as

$$\dot{\tilde{x}} = (A - LC)\tilde{x} + F_{\theta}(x, \hat{\theta}) + G_u(x, u) - F_{\theta}[\hat{x} + l_1(y - C\hat{x}), \hat{\theta}] - G_u[\hat{x} + l_2(y - C\hat{x}), u] + F_{\theta}(x, \tilde{\theta}).$$

Stability analysis of the state observer error system

Theorem 1: UUB State Estimation Error

Provided Assumption 1 and 2 hold, if the control input remains bounded and the system trajectories remain within the compact set $\mathcal{C}$ where the bounds on the derivatives of Y and g are valid, if there exists a constant symmetric positive definite matrix, P , and three observer gains, l_1 , l_2 and L , that satisfy the matrix inequality,

$$\begin{bmatrix} \left((A-LC)^T P + P(A-LC) \right) + 2\alpha P & P - (\mathbb{I}_n - l_1 C)^T (M_y)_{22} & P - (\mathbb{I}_n - l_2 C)^T (M_g)_{22} \\ P - (M_y)_{21} (\mathbb{I}_n - l_1 C) & -(M_y)_{22} & 0 \\ P - (M_g)_{21} (\mathbb{I}_n - l_2 C) & 0 & -(M_g)_{22} \end{bmatrix} < 0,$$

where P is a symmetric positive definite matrix, and M_y and M_g are multiplier matrices designed as

$$M_y = \begin{bmatrix} 0 & -\frac{K_{y2}^T - K_{y1}^T}{2} \\ -\frac{K_{y2} - K_{y1}}{2} & \mathbb{I}_n \end{bmatrix}, \text{ and } M_g = \begin{bmatrix} 0 & -\frac{K_{g2}^T - K_{g1}^T}{2} \\ -\frac{K_{g2} - K_{g1}}{2} & \mathbb{I}_n \end{bmatrix}.$$

then observer error $\tilde{x} = x - \hat{x}$ is locally uniformly ultimately bounded.

Lemma 1

Given a time delay $\varsigma \geq 0$, if $x, \hat{x} \in \mathcal{C}$ and if Assumption 1 holds, for all $\varsigma \geq 0$ and for all $t \geq \varsigma$, the state estimates satisfy $\hat{x}(t) - \hat{x}(t - \varsigma) = \hat{\mathcal{Y}}(t)\theta + \hat{\mathcal{G}}_u(t) + \mathcal{E}(t), \forall t \geq \varsigma$, where $\hat{\mathcal{Y}}(t) := \int_{t-\varsigma}^t Y(\hat{x}(\tau))d\tau$, $\hat{\mathcal{G}}_u(t) := \int_{t-\varsigma}^t g(\hat{x}(\tau))u(\tau)d\tau$, and $\mathcal{E}(t) = O\left(\sup_{\sigma \in [t-\varsigma, t]} \|\tilde{x}(\sigma)\|\right)$.

- Lemma 1 implies that the parameter estimation error at any time t can be expressed as $\hat{\mathcal{Y}}(t)\tilde{\theta}(t) = \hat{x}(t) - \hat{x}(t - \varsigma) - \hat{\mathcal{G}}_u(t) - \hat{\mathcal{Y}}(t)\hat{\theta}(t) - \mathcal{E}(t)$, which motivates the update law

$$\dot{\hat{\theta}} = \begin{cases} k_{\theta}\Gamma\phi, & \text{if } \hat{\theta}^{\top}\hat{\theta} < \bar{\theta}^2 \text{ or if} \\ & \hat{\theta}^{\top}\hat{\theta} = \bar{\theta}^2 \text{ and } (k_{\theta}\Gamma\phi)^{\top}\hat{\theta} \leq 0 \\ \left(\mathbb{I}_p - \frac{\Gamma\hat{\theta}\hat{\theta}^{\top}}{\hat{\theta}^{\top}\Gamma\hat{\theta}}\right) k_{\theta}\Gamma\phi, & \text{otherwise} \end{cases}$$

where

$$\phi(t) := \sum_{i=1}^N \left(\frac{\hat{\mathcal{Y}}(t_i(t))}{1 + \kappa \|\hat{\mathcal{Y}}(t_i(t))\|^2} \right)^{\top} (\hat{x}(t_i(t)) - \hat{x}(t_i(t) - \varsigma) - \hat{\mathcal{G}}_u(t_i(t)) - \hat{\mathcal{Y}}(t_i(t))\hat{\theta}(t)).$$

Parameter Update law and History Stack

- The matrix $\Gamma \in \mathbb{R}^{p \times p}$ is the least-squares gain matrix updated as

$$\dot{\Gamma}(t) = \begin{cases} \beta_1 \Gamma(t) - k_\theta \Gamma(t) \frac{\hat{Y}(t_i(t))^\top \hat{Y}(t_i(t))}{1 + \kappa \|\hat{Y}(t_i(t))\|^2} \Gamma, & \text{if } \hat{\theta}^\top(t) \hat{\theta}(t) < \bar{\theta}^2 \text{ or if} \\ & \hat{\theta}^\top(t) \hat{\theta}(t) = \bar{\theta}^2 \text{ and } (k_\theta \Gamma(t) \phi(t))^\top \hat{\theta}(t) \leq 0 \\ 0 & \text{otherwise} \end{cases}$$

- The update law relies on the time delay ς , and a history stack $\mathcal{H}$, which is the set of piecewise constant functions

$$\hat{\mathcal{X}}(t) := \begin{bmatrix} \hat{x}(t_1(t)) - \hat{x}(t_1(t) - \varsigma) \\ \vdots \\ \hat{x}(t_N(t)) - \hat{x}(t_N(t) - \varsigma) \end{bmatrix}, \hat{\mathcal{Y}} := \begin{bmatrix} \hat{Y}(t_1(t)) \\ \vdots \\ \hat{Y}(t_N(t)) \end{bmatrix}, \hat{\mathcal{G}}_u := \begin{bmatrix} \hat{G}_u(t_1(t)) \\ \vdots \\ \hat{G}_u(t_N(t)) \end{bmatrix}$$

where $\hat{\mathcal{X}}(t) \in \mathbb{R}^{nN}$, $\hat{\mathcal{Y}}(t) \in \mathbb{R}^{nN \times p}$ and $\hat{\mathcal{G}}_u(t) \in \mathbb{R}^{nN}$.

Convergence of Parameter Estimates

- Under Lemma 1, the parameter estimation error dynamics can be expressed as

$$\dot{\tilde{\theta}} = -k_{\theta}\Gamma \underbrace{\sum_{i=1}^N \frac{\hat{\mathcal{Y}}(t_i)^{\top} \hat{\mathcal{Y}}(t_i)}{1 + \kappa \|\hat{\mathcal{Y}}(t_i)\|^2}}_{\Psi} \tilde{\theta} - k_{\theta}\Gamma \underbrace{\sum_{i=1}^N \frac{\hat{\mathcal{Y}}(t_i)^{\top} \mathcal{E}(t_i)}{1 + \kappa \|\hat{\mathcal{Y}}(t_i)\|^2}}_Q.$$

Assumption 3: Eigenvalue Condition

For a given $N \in \mathbb{N}$, there exist a set of time instances $\{t_i\}_{i=1}^N$ such that

$$\lambda_{\min} \left(\sum_{i=1}^N \frac{\hat{\mathcal{Y}}(t_i)^{\top} \hat{\mathcal{Y}}(t_i)}{1 + \kappa \|\hat{\mathcal{Y}}(t_i)\|^2} \right) = \underline{c} > 0.$$

- A history stack that meets the eigenvalue condition in Assumption 3 is called *full rank*.
- Convergence rate depends on the lower bound $\underline{c}$, so a minimum eigenvalue maximization algorithm is utilized for the selection of the piecewise constant time instances $\{t_i(t)\}_{i=1}^N$.

Convergence Results of Parameter Estimator

Definition 1: Finitely Informative (FI) and Persistently Informative (PI) Data

The signal $(\hat{x}, u)$ is called finitely informative (FI) if there exist time instances $0 \leq t_1 < t_2 < \dots < t_N$, for some finite positive integer N , such that the resulting history stack is full rank and persistently informative (PI) if, for any $T \geq 0$, there exist time instances $T \leq t_1 < t_2 < \dots < t_N$ such that the resulting history stack is full rank.

Theorem 2: ISS Parameter Estimation Error

Under the conditions of Theorem 1 and Assumption 3, if the signal $(\hat{x}, u)$ is FI and $\mathcal{H}$ is populated using the minimum eigenvalue maximization algorithm, then the parameter estimation error dynamics are ISS with input $\sqrt{Q}$.

Improving Parameter Estimates using a Greedy Purging Algorithm

- ▶ The initial history stack, recorded during the transient response of the state estimator, may contain inaccurate state estimates.
- ▶ Solution: a purge of the history stack once data corresponding to *more accurate* state estimates are recorded.
- ▶ Exponential decay of state estimation error implies that newer estimates are better.
- ▶ The greedy purging algorithm uses two history stacks:
 - (a) A main stack denoted as $\mathcal{H}$ and a transient stack labeled $\mathcal{G}$.
 - (b) The transient stack is filled until a user-selected dwell time $\mathcal{T} > 0$ has elapsed. Then, the main stack is purged, and the transient stack is copied into the main stack.

Final Comments on convergence of the parameter estimator

- If $(\hat{x}, u)$ is FI with the excitation lasting long enough so that $\mathcal{H}_3$ is full rank, then given any $\varepsilon > 0$

$$\limsup_{t \rightarrow \infty} \|\tilde{\theta}(t)\| \leq \sqrt{\frac{\bar{\Gamma}}{\kappa \underline{\Gamma}}} \left(\frac{2k_{\theta} N L_e}{\underline{a}} \right) \varepsilon.$$

where $\underline{a} := k_{\theta} \underline{c} + \frac{\beta_1}{\bar{\Gamma}}$, $\underline{c}$ is defined in Assumption 3, $L_e \in \mathbb{R}_+$ is a constant such that $\|\mathcal{E}\| \leq L_e \sup_{t \in I_2} \|\tilde{x}(t)\|$.

► System:

$$Y(x) = \begin{bmatrix} (x)_2 & 0 & 0 & 0 \\ 0 & (x)_1 & (x)_2 & x_2(\cos(2(x)_1) + 2)^2 \end{bmatrix}, \quad g(x) = \begin{bmatrix} 0 \\ \cos(2(x)_1) + 2 \end{bmatrix}$$

$$C = [1; 0], \quad \theta = [1; -1; -0.5; 0.5]$$

- Controller: PD controller that tracks the trajectory $(x_d)_1(t) = -\frac{1}{3} \cos(3t) - \frac{1}{2} \cos(2t)$.
- LMI in Theorem 1 is solved using SEDUMI in YALMIP on MATLAB with the learning rate selected as $\alpha = 2$.
- The observer-identifier system is formulated as an event-triggered delay differential equation (DDE), solved using MATLAB's DDE solver, DDE23.

Simulation Results

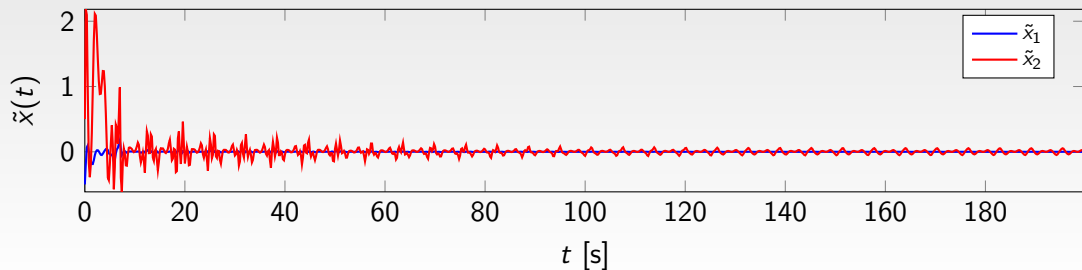


Figure: Trajectory of state estimation error

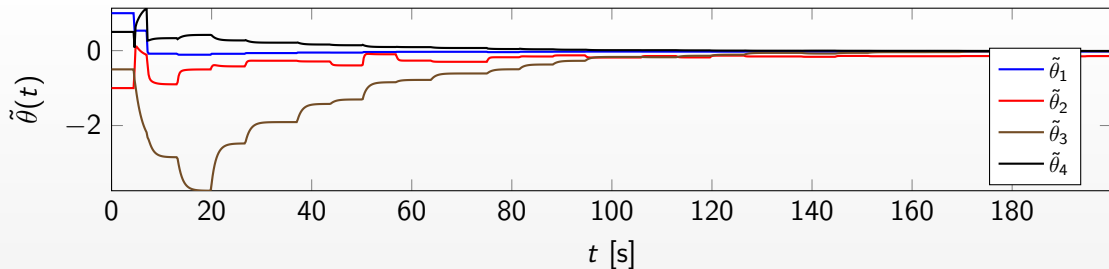


Figure: Trajectory of parameter estimation error

Conclusion

Summary:

- ▶ Developed an online scheme for joint state and parameter estimation in nonlinear systems by integrating multiplier matrix observer design with an event-based implementation of concurrent learning.
- ▶ Developed a greedy purging algorithm to enhance parameter identification by using newer and more accurate state estimates.

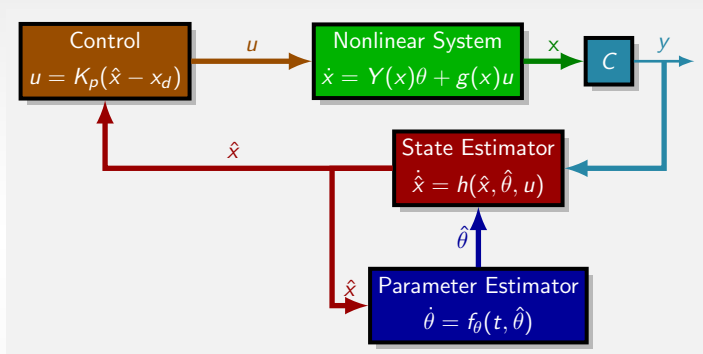
Future work:

- ▶ Analysis of identifiability for nonlinear systems
- ▶ Integration with adaptive optimal control
- ▶ Delay embedding and input-output models for optimal control

Acknowledgements:

- ▶ AFRL FA8651-23-1-0006
- ▶ ONR N00014-21-1-2481

Thank you for your attention!



Questions?